LOCAL_LANGUAGE_DETECTOR_PLAN_v1.md

Implementation plan — sovereign, on-device language detection for QashqAI Voice

Status: Draft v1 — Layer 1 (Claude Code) execution brief **Priority:** #1 (per Desk Mode + Layer 2 verdict, 2026-06-17) **Governance:** No raw user data leaves QashqAI Voice-controlled European infrastructure. Aligns with Doctrine #28/#29 (no LLM in raw-data runtime path) and #31 (substitutability). Encodes community linguistic knowledge → Layer 4 review applies to the Qashqai marker lists and Living Dictionary use.

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1. Objective

Replace the Claude-dependent Language Detector (pipeline Layer A) with a local module that identifies **Qashqai (Darreh-Shuri)**, **Persian**, **German**, and **English** entirely in-process, with no external API call and no data egress. Claude becomes optional *enrichment*, never a hard dependency.

2. Hard constraints

- **Zero egress.** Detection runs synchronously inside the FastAPI backend (Python). No network call to any model provider.
- **Fail toward uncertainty, not toward a confident wrong label.** Low-confidence inputs return `uncertain`, never a forced guess. This mirrors the project's Verified/Likely/Unverified ethic and the Fail-Closed principle.
- **Non-extractive.** Any model artifacts (n-gram tables, marker lists, trained weights) stay in-house; never shipped to or trained by a third party.

3. The real problem (why this is not "just use langdetect")

Four languages, but **three different difficulty classes**:

1. **Script split — trivial, deterministic.** Persian + Qashqai use Perso-Arabic script; German + English use Latin. A Unicode-block check cleanly separates `{fa, qxq}` from `{de, en}`. No model needed.

2. **German vs English — easy, solved.** Latin-script disambiguation. German-specific characters (ä ö ü ß) + function-word frequency + a small off-the-shelf model handle this reliably.
3. **Persian vs Qashqai — HARD, and unique to this project.** Both share the script. **No off-the-shelf detector has a Qashqai label** — it is an endangered, low-resource Oghuz Turkic variety with no standard model coverage. Generic models will misclassify Qashqai as Persian, Azerbaijani, or Turkish. This is the layer that must be **custom-built from QashqAI Voice's own assets**, and it is precisely where the project's curated data (Living Dictionary, 363+ verified entries) becomes a strategic advantage no competitor has.

So the architecture is a **cascade**, not a single classifier.

4. Architecture — detection cascade

```
input text
  │
  ├─[Stage 0] Script detection (deterministic, Unicode blocks)
  │      ├─ Latin-dominant  → Stage 1A (de vs en)
  │      ├─ Arabic-dominant → Stage 1B (fa vs qxq)   ← the hard one
  │      └─ Mixed / neither → flag mixed=true, segment & detect per segment
  │
  ├─[Stage 1A] de vs en
  │      • strong signal: ä ö ü ß → de
  │      • else: local model (fastText lid.176.ftz OR lingua-py) restricted to {de
  │
  ├─[Stage 1B] fa vs qxq  (CUSTOM)
  │      • Qashqai-marker scorer  (Turkic morphology + lexicon)
  │      • Persian-marker scorer  (Indo-European function words + ezafe patterns)
  │      • Living Dictionary lexical anchor (verified qxq tokens)
  │      • char n-gram model trained on qxq corpus vs fa corpus
  │      → weighted score + confidence + abstention threshold
  │
  └─[Stage 2] Confidence calibration + output contract (§7)
```

5. Stage 1B — the Persian/Qashqai discriminator (the core deliverable)

Because labeled Qashqai data is scarce (~363 entries at start), do **not** train a heavy model that will overfit. Use a **hybrid scorer** that degrades honestly:

(a) Qashqai morphological/lexical markers (Turkic, Oghuz — must be Layer 4-

validated):

- Agglutinative suffixes: plural `-lar/-lər`, ablative `-dan/-dən`, locative `-da/-də`, etc.
- Turkic function/pronoun set: `var`, `yox`, `men/mən`, `sen/sən`, `bu`, `o`, `gəl-/gel-` verb stems, vowel-harmony patterns.
- Oghuz-specific lexemes drawn from the Living Dictionary.

(b) Persian markers:

- High-frequency function words: `می`, `که`, `این`, `آن`, `را`, `است/هست`, `و` as connector.
- Ezafe construction patterns and Persian verb morphology.

(c) Living Dictionary lexical anchor: exact/fuzzy match of input tokens against the 363+ verified Qashqai entries → strong positive evidence for `qxq`.

(d) Character n-gram model: train a lightweight Naive Bayes / logistic-regression char-trigram classifier on a Qashqai corpus vs a Persian corpus. Bootstrap the qxq corpus from the Living Dictionary + transcribed Layer 4 recordings (consented).

Combine (a)–(d) into a weighted score with an **abstention band**: if the margin between fa and qxq is below threshold (short input, loanword-dense, code-switched), return `uncertain` with the top-2 candidates. Honest about the small-corpus reality; improves automatically as the corpus grows.

6. Code-switching (real diaspora usage)

Diaspora users mix Persian / German / English / Qashqai in one message. The module must:

- Detect `mixed=true` rather than forcing one global label.
- Optionally return per-segment labels (sentence/clause granularity).
- Never let a mixed input silently collapse to a single confident label.

7. Output contract

```
{
  "language": "qxq | fa | de | en | uncertain",
  "confidence": 0.0–1.0,
  "method": "script | de_en_model | fa_qxq_hybrid | living_dictionary",
  "mixed": false,
  "segments": [{ "text": "...", "language": "...", "confidence": 0.0 }],
```

```
  "candidates": [{ "language": "fa", "score": 0.0 }, { "language": "qxq", "score"
}
```

`method` is the provenance tag required by PIPELINE_RESILIENCE_SPEC_v1 (§6 audit).

8. Tech choices (FastAPI / Python)

- **Stage 0:** pure-Python `unicodedata` / regex over Unicode blocks. Zero dependency.
- **Stage 1A:** `lingua-py` (strong on short text, gives confidence) or `fasttext` with `lid.176.ftz` (compact). Restrict label set to {de, en}.
- **Stage 1B:** custom — `scikit-learn` for the char-trigram model + a hand-built marker module. No external service.
- All models loaded once at app startup; detection is a synchronous in-memory call (sub-millisecond to low-ms).

9. Phasing & milestones

- **Phase 1 (quick win, no governance gate):** Stage 0 script split + Stage 1A (de/en) + Persian baseline. Ships independently, removes most Claude dependency immediately.
- **Phase 2 (the valuable core):** Stage 1B fa/qxq hybrid discriminator using Living Dictionary + marker lists. Requires Layer 4 validation of marker lists.
- **Phase 3:** code-switching/segmentation + confidence calibration against a held-out test set.

10. Evaluation (pre-registered, per project ethic)

- Build a labeled held-out set (Living Dictionary + sample sentences per language, incl. mixed).
- Metrics: per-language precision/recall; **confusion matrix with focus on fa↔qxq**; abstention rate.
- Pre-register acceptance thresholds before measuring. Report honestly; do not tune to the test set.

11. Governance & risk

- **Layer 4 consent:** marker lists and any transcribed-recording corpus use require cultural-validation sign-off (same gate as raw data).
- **Honest limitation:** at 363 entries the fa/qxq layer starts rough; abstention is the safety valve. State this openly — accuracy grows with the corpus, never overclaim.
- **Asset status:** this detector encodes Qashqai linguistic knowledge → it is itself a cultural asset under Doctrine #31. Keep it in-house and version-controlled.

12. Open questions (Siefollah)

1. Can the current Living Dictionary export be used as a bootstrap corpus, and are transcribed Layer 4 recordings available (consented) to enlarge the qxq training set?
2. Phase 1 ships first as a standalone slice — confirm, or build Phases 1–2 as one PR for Layer 4 review together?
3. Hosting: in-process on Render, or self-hosted EU node (Doctrine #31 jurisdictional independence)?